Arista Access Node WFD-ACC-48 QoS Congestion Event Troubleshooting and Resolution SOP

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_1043

Description

The occurrence of QoS congestion events on Arista Access nodes, such as Node ID WFD-ACC-48 located at the Whitefield POP, stems from the interplay of various factors including traffic patterns, buffer pool utilization, and queue management protocols. In this specific failure domain, microbursts, interface tail drops, and QoS queue starvation are indicative of a mismatch between the configured Quality of Service (QoS) policies and the actual traffic characteristics. The Arista EOS (Extensible Operating System) relies on the P4 (Programming Protocol-Independent Packet Processors) model to execute QoS policies. However, during periods of high network congestion, the interface buffer pools may become overwhelmed, leading to tail drop events and subsequent QoS queue starvation. This scenario is exacerbated by the lack of sufficient buffer credits, which are allocated based on the configured QoS policies and the traffic characteristics. As a result, the Arista access node WFD-ACC-48 may exhibit symptoms such as packet drops, increased latency, and degradation of network services.

Pre-requisite Checks

- check: Verify the network topology and ensure that there are no physical layer issues or hardware failures affecting the Arista access node WFD-ACC-48.

- check: Confirm that the redundant link status is active and functioning correctly.

- check: Ensure that the necessary software licenses and feature sets are enabled on the Arista access node WFD-ACC-48.

Potential Root Causes

Cause	Probability
Inadequate buffer pool allocation due to misconfigured QoS policies.	High
Insufficient buffer credits allocated to QoS queues.	Medium
Interface tail drop events due to excessive buffer utilization.	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface Status: Up/Down, Protocol: up/up, Description: This command verifies the interface status and protocol state, helping to identify any potential hardware or software issues affecting the Arista access node WFD-ACC-48.
```

```
Command: show qos policy statistics, Expected Output: QoS Policy Statistics: Packets Matched, Packets Dropped, Description: This command provides detailed statistics on QoS policy matches, drops, and other relevant metrics, enabling the identification of potential QoS policy configuration issues or mismatches.

Command: show buffer credits, Expected Output: Buffer Credits: Allocated, Used, Available, Description: This command displays the allocated, used, and available buffer credits for each interface, helping to identify potential buffer pool allocation issues or mismatches.
```

Resolution Steps

Step 1: Step: Reconfigure QoS policies to optimize buffer pool allocation., Command: config qos policy buffer-credits , Description: This step involves reconfiguring the QoS policies to allocate sufficient buffer credits for each interface, ensuring optimal buffer pool utilization and minimizing the occurrence of tail drop events.

Step 2: Step: Increase buffer credits allocated to QoS queues., Command: config qos queue buffer-credits , Description: This step involves increasing the buffer credits allocated to each QoS queue, ensuring sufficient buffer credits for optimal queue management and minimizing the occurrence of queue starvation events.

Step 3: Step: Verify and apply the latest software release with QoS enhancements., Command: upgrade system software , Description: This step involves verifying and applying the latest software release with enhanced QoS features and bug fixes, ensuring optimal QoS performance and minimizing the occurrence of QoS-related issues.

Post-Check Verification

```
Command: show qos policy statistics, Expected Output: QoS Policy Statistics: Packets Matched, Packets Dropped, Description: This command verifies that QoS policy matches and drops have been optimized, ensuring that the reconfigured QoS policies are effective in managing traffic and minimizing congestion events.

Command: show buffer credits, Expected Output: Buffer Credits: Allocated, Used, Available, Description: This command verifies that buffer credits have been allocated and utilized optimally, ensuring that the reconfigured QoS policies are effective in managing buffer pool utilization and minimizing congestion events.
```

Roll-back Procedures

- Step: Revert QoS policies to previous configuration., Command: config qos policy rollback, Description: This step involves reverting the QoS policies to their previous configuration, ensuring that any changes made during the troubleshooting process do not impact network performance or functionality.

- Step: Restore buffer credits to previous values., Command: config qos queue buffer-credits , Description: This step involves restoring the buffer credits to their previous values, ensuring that any changes made during the troubleshooting process do not impact buffer pool utilization or network performance.

- Step: Revert software release to previous version., Command: rollback system software , Description: This step involves reverting the software release to its previous version, ensuring that any changes made during the troubleshooting process do not impact network performance or functionality.